```python
#Task 1

import librosa as lb
y,sr=lb.load("C:\\Users\\krsub\\OneDrive\\Desktop\\internship\\game_over.wav", sr=None)
print("Audio duration: ",lb.get_duration(y=y, sr=sr),"seconds")
print("Sampling Rate:", sr,"Hz")
import sounddevice as sd
sd.play(y, sr)
sd.wait()


#Task 2

import matplotlib.pyplot as plt
import librosa.display
plt.figure(figsize=(10,4))
lb.display.waveshow(y,sr=sr)
plt.title("Waveform")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.show()

#Task 3
import numpy as np
d=lb.amplitude_to_db(np.abs(lb.stft(y)),ref=np.max)
plt.figure(figsize=(10,4))
lb.display.specshow(d,sr=sr,x_axis="time",y_axis="log")
plt.title("Spectrogram")
plt.show()

#Task 4
mfccs=lb.feature.mfcc(y=y,sr=sr,n_mfcc=13)
plt.figure(figsize=(10,4))
lb.display.specshow(mfccs,sr=sr,x_axis="time")
plt.title("MFCCs")
plt.show()

#Task 5
chroma=lb.feature.chroma_stft(y=y,sr=sr)
plt.figure(figsize=(10,4))
lb.display.specshow(chroma,sr=sr,x_axis="time",y_axis="chroma")
plt.title("Chroma features")
plt.show()
```

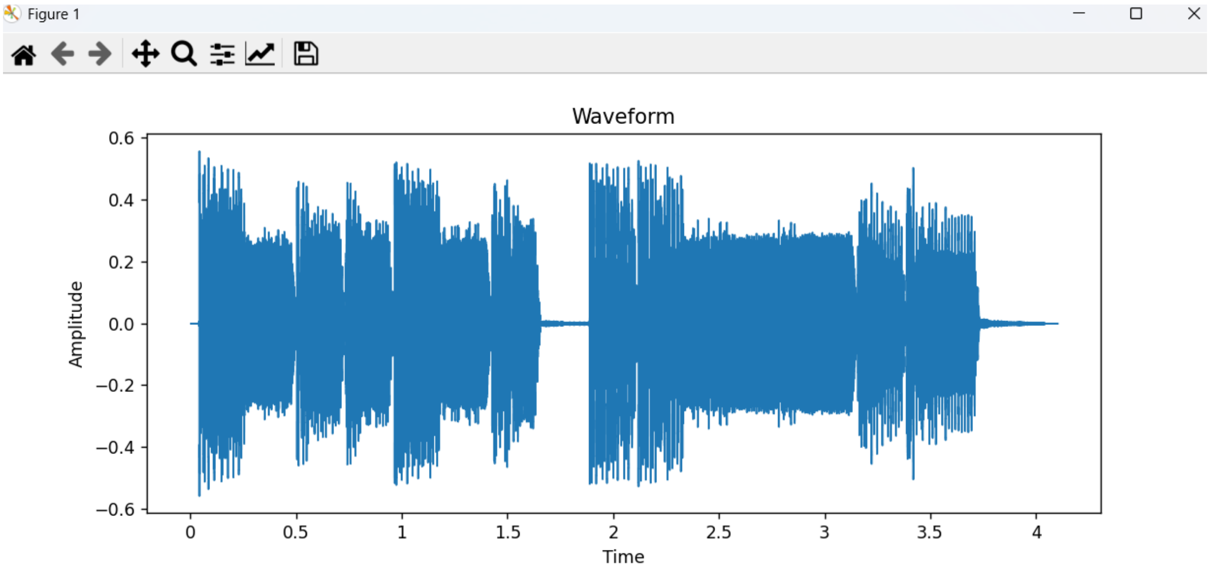
```python
#Task 6
tempo, beats=lb.beat.beat_track(y=y,sr=sr)
print("Estimated tempo:",tempo,"BPM")
beat_times=lb.frames_to_time(beats,sr=sr)
plt.figure(figsize=(10,4))
lb.display.waveshow(y,sr=sr)
plt.vlines(beat_times,ymin=-1,ymax=1,color='r',linestyle='--',label="Beats")
plt.legend()
plt.title("Beat tracking")
plt.show()

#Task 7
z=lb.feature.zero_crossing_rate(y)
plt.figure(figsize=(10,4))
plt.plot(z[0],label="Zero crossing rate")
plt.xlabel("frames")
plt.ylabel("rate")
plt.title("zero crossing rate")
plt.legend()
plt.show()
```

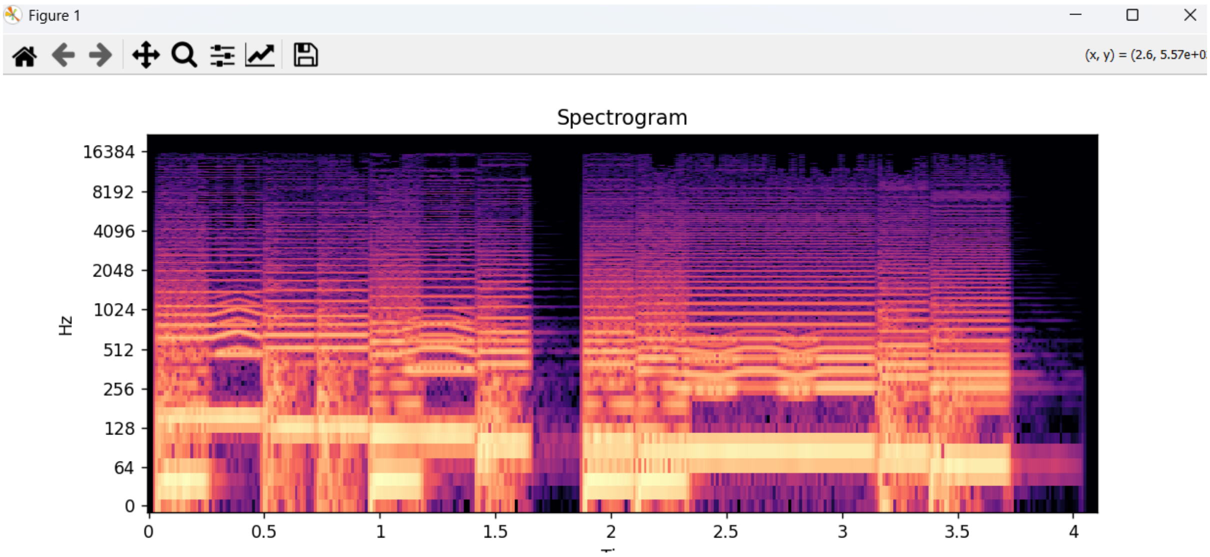
#Task 2

```
========== RESTART: C:/Users/krsub/OneDrive/Desktop/inter
Audio duration:  4.101224489795919 seconds
Sampling Rate: 44100 Hz
```

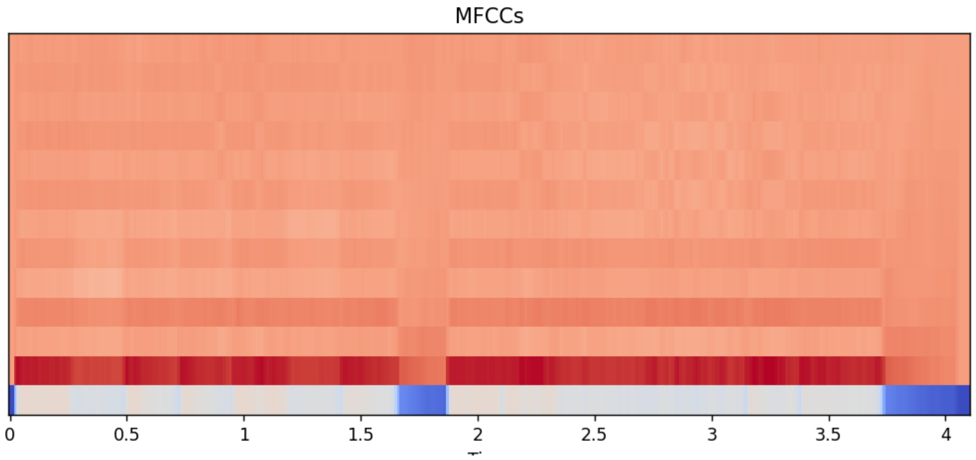
#Task 2



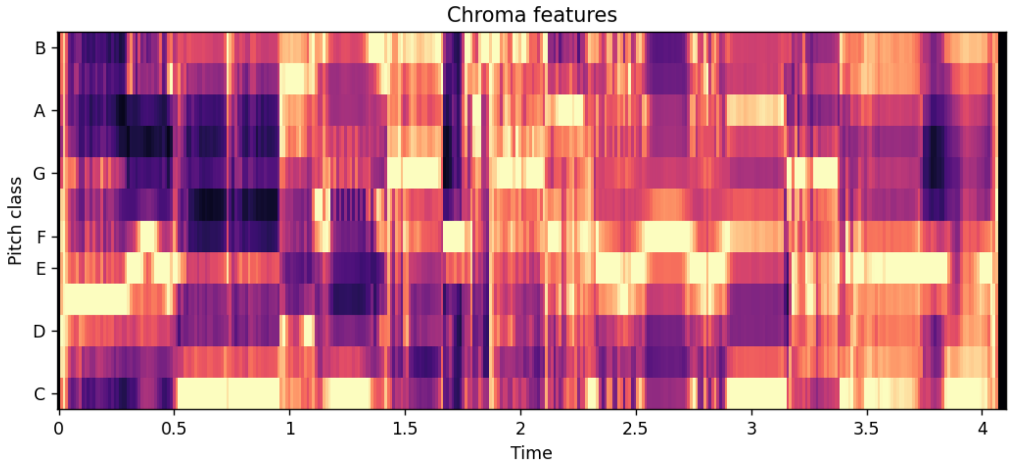
#Task 3



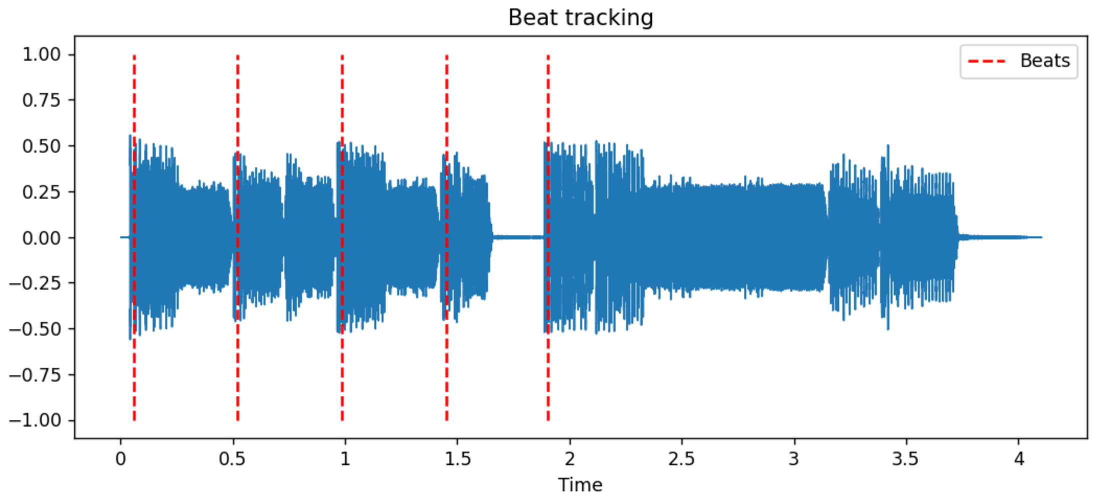
#Task 4



#Task 5

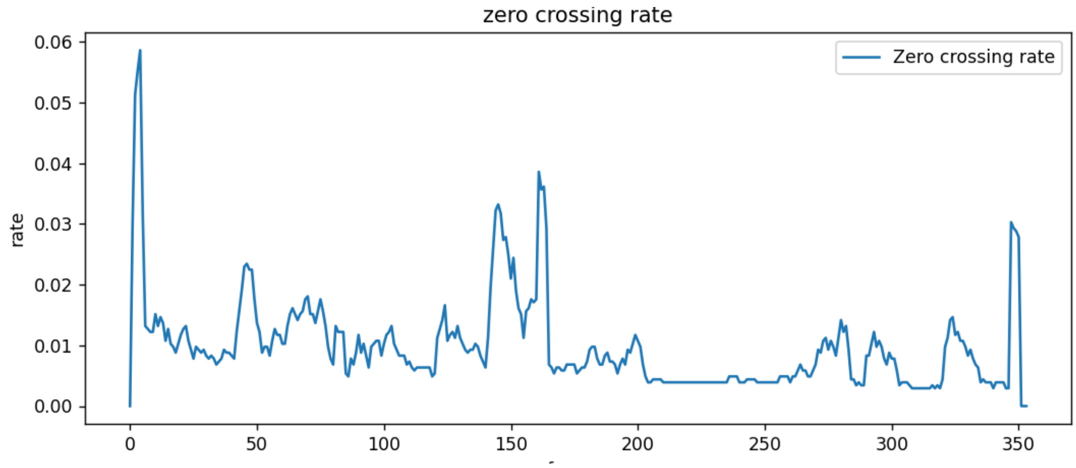


#Task 6




```
Estimated tempo: [135.99917763] BPM
```

#Task 7


zero crossing rate
Zero crossing rate
rate
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
50
100
150
200
250
300
350